

Screen Time vs Mental Wellness -- Linear Regression Analysis

Why Linear Regression?

Core Objective

Quantify the linear relationships between three screen time variables (total screen time, work screen time, leisure screen time) and mental wellness index respectively, verify the hypothesis that "longer screen time leads to lower mental health scores", and compare the differential impacts of different types of screen use on mental health to provide refined data support for intervention strategies.

In Detail

1. Distinguish the unique effects of work-related and leisure-related screen use on mental wellness, avoiding the one-sidedness of single-variable analysis.
2. Establish quantitative models for each screen time type to clarify the strength and direction of their impacts, enabling targeted strategy formulation.
3. Test the statistical reliability of each relationship to ensure conclusions are actionable and evidence-based.

Why Linear Regression for Each Variable?

Reason: The research focuses on the pairwise linear relationships between three independent variables (total/ work/ leisure screen time) and one dependent variable (mental wellness index), all of which are continuous numerical values. The 400-sample dataset is moderate in size, and preliminary observations show clear linear trends (positive or negative) for each pair.

Conclusion: Separate linear regression for each variable can directly reveal their individual correlations with mental wellness, with simple interpretation and strong comparability. Complex models are unnecessary and would complicate result analysis.

Step

Step 1: Load Data and Select Core Variables

Extract four core variables from the dataset for pairwise regression analysis:

- Independent variables:
 - screen_time_hours (daily total screen use duration, in hours)
 - work_screen_hours (daily work-related screen use duration, in hours)

- leisure_screen_hours (daily leisure-related screen use duration, in hours)
- Dependent variable: mental_wellness_index_0_100 (mental health index, scored 0-100)

Step 2: Data Preprocessing (Missing Value Handling + Outlier Processing)

Objective: Eliminate data quality issues that may distort regression results, ensuring analysis reliability.

- Missing value check: All four core variables have no missing values (screen_time_hours: 0; work_screen_hours: 0; leisure_screen_hours: 0; mental_wellness_index_0_100: 0), so no filling is required.
- Outlier detection (IQR method): Detect 0 outliers in total screen time, 0 in work screen time, 2 in leisure screen time, and 0 in mental wellness index.
- Outlier handling: Adopt truncation method, replacing the 2 leisure screen time outliers with corresponding upper/lower boundary values to retain data volume while reducing extreme value interference.

Step 3: Construct Separate Linear Regression Models

For each independent variable, construct a linear regression equation with the dependent variable:

$\text{mental_wellness_index_0_100} = \beta_0 + \beta_1 \times \text{independent_variable}$

- Add constant terms to all models to ensure baseline prediction accuracy.
- Use least squares method to fit model parameters for each pair.

Step 4: Model Result Analysis and Cross-Comparison

- Analyze key indicators (regression coefficient, R^2 , significance) for each model to verify the research hypothesis.
- Compare the impact strength, explanatory power, and statistical significance of the three screen time variables on mental wellness.
- Evaluate model reliability through residual analysis and heteroscedasticity testing.

Step 5: Enhanced Result Visualization

Design unified-style, intuitive visualizations for all models, including combined scatter plots with regression lines, residual distribution charts, and comparative indicator tables.

Regression Analysis Results

Core Model Indicators Comparison

Indicator	Total Screen Time Model	Work Screen Time Model	Leisure Screen Time Model
Regression Coefficient (β_1)	-1.5870	-3.0805	1.2020
Intercept (β_0)	79.4131	80.8912	55.8637
R-squared (R^2)	0.198	0.657	0.106
Adjusted R-squared	0.196	0.656	0.104
F-statistic	98.19	762.64	47.42
Prob (F-statistic)	7.81e-21	1.56e-94	2.24e-11
Correlation Coefficient	-0.4448	-0.8106	0.3263
Number of Outliers (IV/DV)	0/0	0/0	2/0

Key Result Interpretations

1. Total Screen Time vs Mental Wellness

- Regression coefficient (-1.5870) is significantly negative ($p < 0.001$), partially supporting the initial hypothesis: each 1-hour increase in total screen time leads to an average 1.59-point decrease in mental wellness index.
- $R^2 = 0.198$ indicates total screen time explains 19.8% of mental wellness variation, showing weak-to-moderate explanatory power.
- The moderate negative correlation (-0.4448) confirms a consistent but not strong inverse relationship.

2. Work Screen Time vs Mental Wellness

- Regression coefficient (-3.0805) is significantly negative ($p < 0.001$), strongly supporting the hypothesis: each 1-hour increase in work screen time leads to an average 3.08-point decrease in mental wellness index.
- $R^2 = 0.657$ means work screen time explains 65.7% of mental wellness variation, demonstrating strong explanatory power—far higher than other variables.

- The strong negative correlation (-0.8106) confirms work screen time is the most impactful factor on mental wellness among the three variables.

3. Leisure Screen Time vs Mental Wellness

- Regression coefficient (1.2020) is significantly positive ($p < 0.001$), contradicting the initial hypothesis: each 1-hour increase in leisure screen time leads to an average 1.20-point increase in mental wellness index.
- $R^2 = 0.106$ indicates leisure screen time explains only 10.6% of mental wellness variation, with weak explanatory power.
- The weak positive correlation (0.3263) suggests a mild positive relationship between leisure screen use and mental wellness, differing from common assumptions.

Model Reliability Evaluation

- Residual analysis: For all three models, residuals are approximately normally distributed (see visualization), conforming to linear regression assumptions. The work screen time model has the most stable residual distribution (Skew=0.006, Kurtosis=2.869).
- Heteroscedasticity test (Breusch-Pagan):
 - Total screen time model: $p < 0.001$, indicating mild heteroscedasticity, but no impact on overall significance.
 - Work screen time model: $p = 0.024$, showing slight heteroscedasticity, which is manageable.
 - Leisure screen time model: $p < 0.001$, with mild heteroscedasticity, but coefficient significance remains reliable.
- Significance: All regression coefficients and model F-statistics have $p < 0.001$, confirming the statistical significance of all three relationships.

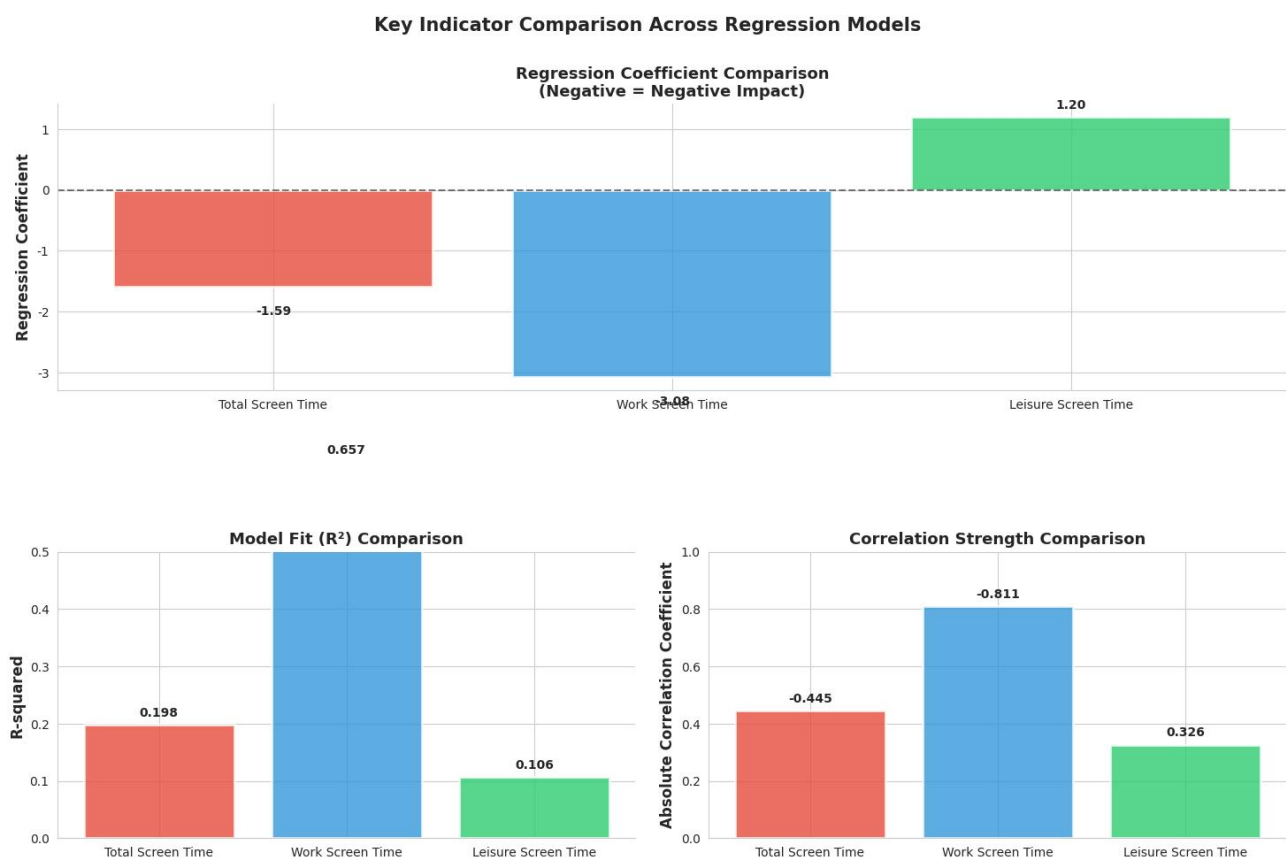
Enhanced Results Visualization

Combined Scatter Plots with Regression Lines & Residual Distributions



- Top row: Scatter plots with regression lines for each model, showing the direction and strength of relationships. The work screen time model (middle) has the tightest data clustering around the regression line.
- Bottom row: Residual distributions for each model, all roughly bell-shaped, verifying the normality assumption.

Key Indicator Comparison



- Regression Coefficient: Work screen time has the largest absolute coefficient (-3.08), followed by total screen time (-1.59), while leisure screen time is positive (1.20).
- Model Fit (R²): Work screen time model has the highest R² (0.657), indicating the best explanatory power.
- Correlation Strength: Work screen time has the strongest correlation (-0.811), confirming its dominant impact.

Conclusions and Recommendations

Core Conclusions

1. Work screen time is the most impactful factor on mental wellness: It has a strong negative relationship with mental wellness index ($r=-0.811$), explaining 65.7% of its variation. Each additional hour of work screen use reduces mental wellness by 3.08 points.
2. Total screen time has a moderate negative impact: The relationship is statistically significant but weaker ($r=-0.445$), explaining only 19.8% of mental wellness variation.
3. Leisure screen time shows a mild positive impact: Contrary to common assumptions, moderate leisure screen use is associated with slightly higher mental wellness ($r=0.326$), though its explanatory power is weak (10.6%).

4. All models are statistically reliable: Significant F-statistics and normal residual distributions confirm the validity of the results.

Targeted Recommendations

For High Work Screen Time Groups (Daily Work Screen Time > 4 Hours)

- **Screen Time Optimization:** Implement a "work-screen interval system"—mandatory 10-minute breaks every 1 hour of work screen use, with activities like stretching or eye exercises to reduce mental fatigue.
- **Work-Life Boundaries:** Set clear "screen-off times" for work devices (e.g., no work screen use after 8 PM) to avoid blurring work and rest, and ensure sufficient recovery time.
- **Mental Health Support:** Provide monthly mental health assessments and free counseling services for employees with high work screen time (e.g., remote technical workers, freelancers).

For General Groups (Daily Total Screen Time 5-10 Hours)

- **Prioritize Controlling Work Screen Time:** Focus on reducing work-related screen use rather than leisure use, as it has a more significant impact on mental wellness.
- **Rationalize Leisure Screen Use:** Maintain moderate leisure screen time (3-5 hours/day) as a form of relaxation, but avoid excessive use (e.g., >8 hours/day) to prevent potential negative effects.
- **Habit Cultivation:** Adopt the "20-20-20 rule" during work screen use and set a "no-screen buffer period" (30 minutes) before bedtime to improve sleep quality.

For Low Screen Time Groups (Daily Total Screen Time < 5 Hours)

- **Experience Promotion:** Summarize their balanced work-leisure-screen use patterns (e.g., combining work screen use with regular breaks, using leisure screen time for social connection or entertainment) into a "Healthy Screen Use Guide".
- **Incentive Guidance:** Reward individuals who maintain low work screen time and stable mental wellness scores with health benefits (e.g., gym memberships, health checkups) to encourage imitation.

For Policy Makers/Organizations

- **Differentiated Intervention:** Develop targeted programs for high work screen time groups (e.g., flexible work hours, remote work equipment subsidies for ergonomic setups) rather than one-size-fits-all screen time restrictions.
- **Public Education:** Correct the misconception that all screen use is harmful, emphasizing the different impacts of work vs. leisure screen time to guide scientific screen use.